

Hridoy Sankar Dutta

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LINKS

Website: <https://hridaydutta123.github.io>
Github: <https://github.com/hridaydutta123>
DBLP: <https://tinyurl.com/hsd-dblp>
Google Scholar: <https://tinyurl.com/hsd-google-scholar>
LinkedIn: <https://tinyurl.com/hsd-linkedin>

WORK EXPERIENCE

Lecturer July 2024 - Present
*Faculty of Science Engineering and Built Environment/School of Information Technology
Deakin University*

Research Associate February 2021 - June 2024
*Department of Computer Science and Technology & Cambridge Cybercrime Centre
University of Cambridge*

Assistant Project Engineer August 2014 - April 2015
IIT Guwahati Project Title: Developing Text to Speech System in Assamese Language Phase II
Supervisor: Prof. S.R.M.Prasanna, Professor, Dept. of EEE, Indian Institute of Technology, Guwahati (IITG)

EDUCATION

PhD in Computer Science and Engineering July 2017 - September 2021
Indraprastha Institute of Information Technology Delhi (IIIT-D)

- Thesis Title: *Detecting and reasoning collusive activities in online media*
- Supervisor: **Dr. Tanmoy Chakraborty**, Associate Professor & Ramanujan Fellow, Dept. of CSE, Indraprastha Institute of Information Technology Delhi (IIIT-D), India

M.Tech in Computer Science and Engineering 2015 - 2017
National Institute of Technology Durgapur, West Bengal, India

- Thesis Title: *An offline crisis mapping system for large-scale natural disasters*
- Supervisor: **Prof. Subrata Nandi**, Professor, Dept. of CSE, NIT Durgapur and **Dr. Sujoy Saha**, Assistant Professor, Dept. of CSE, NIT Durgapur
- CGPA of **9.15/10** (Ranked 3rd)

B.Tech in Computer Science and Engineering 2009 - 2013
Institute of Science and Technology, Gauhati University, Assam, India

- Thesis Title: *Hierarchical Convergecasting with clustering and spanning tree generation*
- Supervisor: **Dr. Sushanta Karmakar**, Associate Professor, Dept. of CSE, Indian Institute of Technology Guwahati (IIT-G)
- Mentor: **Dr. Sandip Chakraborty**, Associate Professor, Dept. of CSE, Indian Institute of Technology Kharagpur (IIT-KGP)
- CGPA of **9.17/10**

PATENT

- Sujoy Saha, Prithviraj Pramanik, Partha Sarathi Paul, Subrata Nandi, Kingshuk De, Bishak Chandra Ghosh, **Hridoy Sankar Dutta**, Tamal Mondal, Krishnendu Hazra, Indrajit Bhattacharya, Sandip Chakraborty, An End-To-End System For Offline Localized Crisis Mapping And Method Of Operation Thereof, Indian Patent, Application No: 201931050685, Filed on 09 December, 2019.

ACCEPTED / PUBLISHED PAPERS

([Top conferences and high impact journals are highlighted in blue.](#))
* - Equal contribution

- [**InterSpeech-26**] Kamel Kamel, **Hridoy Sankar Dutta**, Keshav Sood and Sunil Aryal. Spectral Masking and Interpolation Attack (SMIA): A Black-box Adversarial Attack against Voice Authentication and Anti-Spoofing Systems. [27th Conference of the International Speech Communication Association \(InterSpeech 2026\)](#)., September 27 – October 1, 2026.

- [SIGIR-26] **Hridoy Sankar Dutta**, Biswadeep Khan and Amit Nanavati. GONets: A First-Look Into GitHub Organisation Networks. *49th International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR 2026)*, July 20 – 24, 2026.
- [SIGIRW-25] **Hridoy Sankar Dutta**, Sana Ansari, Swati Kumari and Shounak Ravi Bhalerao. Open-CoderRank: Personalized Technical Assessments with Generative AI. *The First Workshop on Synthetic Content in Information Retrieval Ecosystems, 49th International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR 2026)*, July 20 – 24, 2026.
- [IEEE-IoT] Kamel Kamel, Keshav Sood, **Hridoy Sankar Dutta** and Sunil Aryal. Rethinking Targeted Data Poisoning in Voice Authentication: A Critique and Defense Mechanism, *IEEE Internet of Things Journal*.
- [WiSec-26] Luis Adán Saavedra del Toro, **Hridoy Sankar Dutta**, Alastair Beresford and Alice Hutchings. iOSZoo: A Large-Scale Study of Third-Party iOS App Markets. *ACM Conference on Security and Privacy in Wireless and Mobile Networks (ACM WiSec 2026)*, June 30 – July 3, 2026, Saarbrücken, Germany.
- [CIKMW-25] Swati Kumari, **Hridoy Sankar Dutta** and Hitesh Tiwari. Securing User Identity in Web3 Social Platforms: A Post-Quantum Biometric Approach. *6th Workshop on Secure IoT, Edge and Cloud systems, The 34th ACM Conference on Information and Knowledge Management (CIKM 2025)*, Nov 10 – 14, 2025.
- [CIKM-25] **Hridoy Sankar Dutta** and Biswadeep Khan. YTCommentVerse: A Multi-Category Multi-Lingual YouTube Comment Corpus. *34th ACM Conference on Information and Knowledge Management (CIKM 2025)*, Nov 10 – 14, 2025.
- **Hridoy Sankar Dutta** and Yanna Papadodimitraki. Backlinks for Sale: A First Look into Marketplaces Selling Wikipedia Backlinks *12th Annual Wiki Workshop (WikiWorkshop 2025)*, May 21 – 22, 2025.
- [ASIACCS-25] Luis Adán Saavedra del Toro, **Hridoy Sankar Dutta**, Alastair Beresford and Alice Hutchings. App-solutely Modded: The rift between modded app market operators and original developers. *ACM ASIA Conference on Computer and Communications Security (ACM ASIACCS 2025)*, August 25 – 29, 2025, Ha Noi, Vietnam.
- [LOD-25] Anahit Sargsyan, **Hridoy Sankar Dutta** and Juergen Pfeffer. Half-life of Youtube News Videos: Diffusion Dynamics and Predictive Factors. *The 11th International Conference on Machine Learning, Optimization, and Data Science (LOD 2025)*, September 21 – 24, 2025, Tuscany, Italy.
- [eCrime-24] Luis Adán Saavedra del Toro, **Hridoy Sankar Dutta**, Alastair Beresford and Alice Hutchings. ModZoo: A Large-Scale Study of Modded Android Apps and their Markets, *APWG Symposium on Electronic Crime Research (eCrime 2024)*, September 24 – 26, 2024, Boston, Massachusetts, USA.
- Quincy Taylor, Derek Hansen, **Hridoy Sankar Dutta** and Justin Giboney. Analyzing online media platforms for hacktivist group organization and proliferation, *The 15th Dewald Roode Workshop on Information Systems Security Research (DRW 2023)* 22-23 June 2023, Glasgow, Scotland.
- Luis Adán Saavedra del Toro, Alastair Beresford, **Hridoy Sankar Dutta** and Alice Hutchings. Analysis of Modded Android Apps and Marketplaces, *Fourth UK Mobile, Wearable and Ubiquitous Systems Research Symposium (MobiUK)* 4th-5th July 2022, UCL.
- [ACM TDS] **Hridoy Sankar Dutta** and Tanmoy Chakraborty. Blackmarket-driven collusion on online media: a survey. *ACM/IMS Transactions on Data Science (TDS)* 2, no. 4 (2022): 1-37.
- [ICWSM-22] **Hridoy Sankar Dutta**, Nirav Diwan, Tanmoy Chakraborty. Weakening the Inner Strength: Spotting Core Collusive Users in YouTube Blackmarket Network, *16th International AAAI Conference on Web and Social Media (ICWSM)*. vol. 16, pp. 147-158. 2022.
- [ACM TIST] **Hridoy Sankar Dutta**, Mayank Jobanputra, Himani Negi, Tanmoy Chakraborty. Detecting and Analyzing Collusive Entities on YouTube, *ACM Transactions on Intelligent Systems and Technology (TIST)* 12, no. 5 (2021): 1-28.
- [HyperText-21] **Hridoy Sankar Dutta**, Kartik Aggarwal and Tanmoy Chakraborty. DECIFE: Detecting Collusive Users Involved in Blackmarket Following Services on Twitter, *32nd ACM Conference on Hypertext and Social Media (HyperText)*, Trinity College Dublin, Ireland, pp. 91-100. 2021.
- [ICWSM-21] **Hridoy Sankar Dutta**, Udit Arora and Tanmoy Chakraborty. ABOME: A Multi-platform Data Repository of Artificially Boosted Online Media Entities, *15th International AAAI Conference on Web and Social Media (ICWSM)*, vol. 15, pp. 1000-1008. 2021.

- **Hridoy Sankar Dutta** and Tanmoy Chakraborty. Adversarial Collusion on the Web: State-of-the-Art and Future Directions. In Big Data Analytics: 8th International Conference, BDA 2020, Sonapat, India, December 15–18, 2020, Proceedings 8, pp. 230-235. Springer International Publishing, 2020.
- [ACM TIST] Udit Arora*, **Hridoy Sankar Dutta***, Brihi Joshi, Aditya Chetan and Tanmoy Chakraborty. Analyzing and detecting collusive users involved in blackmarket retweeting activities. *ACM Transactions on Intelligent Systems and Technology (TIST)* 11, no. 3 (2020): 1-24.
- [IEEE TIFS] **Hridoy Sankar Dutta**, Vishal Raj Dutta, Aditya Adhikary and Tanmoy Chakraborty. HawkesEye: Detecting fake retweeters using Hawkes process and topic modeling. *IEEE Transactions on Information Forensics and Security (TIFS)* 15 (2020): 2667-2678.
- [IEEE TIFS] **Hridoy Sankar Dutta** and Tanmoy Chakraborty. Blackmarket-driven Collusion among Retweeters – Analysis, Detection and Characterization. *IEEE Transactions on Information Forensics and Security (TIFS)* 15 (2019): 1935-1944.
- [JNCA] Partha Sarathi Paul, Bishakh Chandra Ghosh, **Hridoy Sankar Dutta**, Kingshuk De, Arka Prava Basu, Prithviraj Pramanik, Sujoy Saha, Sandip Chakraborty, Niloy Ganguly, Subrata Nandi. CRIMP: Here Crisis Mapping Goes Offline. *Elsevier Journal of Network and Computer Applications (JNCA)* 146 (2019): 102418.
- [WSDM-19] Aditya Chetan*, Brihi Joshi*, **Hridoy Sankar Dutta***, Tanmoy Chakraborty. CoReRank: Ranking to Detect Users Involved in Blackmarket-based Collusive Retweeting Activities. *12th ACM International Conference on Web Search and Data Mining (WSDM)*, pp. 330-338. 2019.
- [ASONAM-18] **Hridoy Sankar Dutta**, Aditya Chetan, Brihi Joshi, Tanmoy Chakraborty. Retweet Us, We Will Retweet You: Spotting Collusive Retweeters Involved in Blackmarket Services, *IEEE/ACM International Conference on Advances in Social Networks Analysis and Mining (ASONAM)*, pp. 242-249. IEEE, 2018.
- Partha Sarathi Paul, **Hridoy Sankar Dutta**, Bishakh Chandra Ghosh, Krishnandu Hazra, Sandip Chakraborty, Sujoy Saha, Subrata Nandi. Offline crisis mapping by opportunistic dissemination of crisis data after large-scale disasters, *2nd ACM SIGSPATIAL International Workshop on the Use of GIS in Emergency Management*, pp. 1-8. 2016.
- [SAC-14] Sandip Chakraborty, Suchetana Chakraborty, Sushanta Karmakar, **Hridoy Sankar Dutta**. Hierarchical topology adaptation for distributed convergecast applications, In *29th Annual ACM Symposium on Applied Computing*, pp. 405-407. 2014.

DATASETS

- **Hridoy Sankar Dutta**. YT-30M: A multi-lingual multi-category dataset of YouTube comments. *HuggingFace downloads: 700, Industry adoption: ActiveFence, USA*
- **Hridoy Sankar Dutta**, Udit Arora and Tanmoy Chakraborty . ABOME: A Multi-platform Data Repository of Artificially Boosted Online Media Entities. *Zenodo downloads: 303*

FUNDING

Deakin Cyber Collaboration Grants Scheme 2026 (Co-Pi) <i>Deakin University</i>	6000 AUD
SEBE 2026 Academic Travel Scheme <i>Deakin University</i>	3430 AUD
Peer-review ECR Support Scheme (PRESS) <i>Deakin University</i>	4250 AUD
Cohere Labs Catalyst Grant <i>Cohere Labs</i>	500 USD
School of IT Minor Equipment & Expenses Scheme 2025 <i>Deakin University</i>	3500 AUD
Deakin Cyber Collaboration Grants Scheme 2025 (Co-PI) <i>Deakin Cyber Research and Innovation Centre</i>	5000 AUD 2025 - Present

	OpenAI Researcher Access Program (PI) <i>OpenAI</i>	1000 USD 2025 - Present
	Deakin University Startup Grant (PI) <i>Deakin University</i>	INR 10000 AUD 2024 - Present
TEACHING	Instructor, Deakin University <i>SIT736: Identity Access Management and Physical Security</i> <i>SIT719: Analytics For Security And Privacy</i> <i>SIT763: Cyber Security Management</i> <i>SIT738: Secure Coding</i> <i>SIT753: Professional Practices in IT (SEBE Teaching Excellence Award)</i>	Mar - June 2025 Mar - June 2025 Nov - Jan 2025 July - Oct 2024/2025 July - Oct 2024/2025
	Course Supervisor and Demonstrator, The University of Cambridge <i>Machine Learning and Real-world Data</i> <i>Pembroke Cambridge Summer Programme</i> <i>Security</i>	Jan - Mar 2022, Jan - Mar 2023 July - Aug 2022 May - June 2021
	Teaching Assistant, IIIT Delhi <i>Information Retrieval</i> <i>System Management</i>	Jan - June 2020 August - Nov 2017
	Teaching Assistant, NIT Durgapur <i>Network and Distributed System Laboratory</i> <i>Software Engineering Laboratory</i>	Jan - June 2017 July - Nov 2016
OTHER EXPERIENCES	Co-founder, Assam AI Initiative (AAIL) <i>A non-profit initiative to promote various scopes of AI</i>	Feb 2021 - Present
	Visiting Researcher, TU Munich, Germany <i>Supervisor: Prof. Juergen Pfeffer</i> <i>Professor of Computational Social Science & Big Data, TU Munich</i>	July 2023
	Visiting Researcher, Cardiff University, UK <i>Supervisor: Prof. Pete Burnap</i> <i>Professor, School of Computer Science and Informatics, Cardiff University</i>	March 2019, March 2020
	Short-Term Scholar, George Mason University, USA <i>Supervisor: Dr. Hemant Purohit, Associate Professor, Information Sciences & Technology Department</i> <i>George Mason University (GMU)</i>	June 2019 - August 2019
SELECTED PROJECTS	All projects are available on git : https://www.github.com/hridaydutta123 • The YouTube Scraper - Download YouTube video description and video comments without using the YouTube API. • Assamese2Vec - An open source project for modeling the Assamese language. • Awesome Twitter Tools - A curated list of tools, datasets, research papers and browser extensions related to Twitter. • Collusive User Detection Tool - A tool for detection of collusive users in online social networks. • Opportunistic Network for Android Application - Implementation of communication medium for challenge network scenario. • Offline Master Control Station - Implementation of reliable and effective data transfer from opportunistic network and tool for analyzing behaviour of data captured in opportunistic networks. • Designing TTS System in Assamese and Manipuri Languages - It involves building systems for synthesizing natural speech in native Assamese and Manipuri tone from a given text and building user friendly GUIs for the common people.	

- ACHIEVEMENTS AND AWARDS
- Fellow of the Higher Education Academy (FHEA).
 - SEBE Teaching Excellence Award 2025, Deakin University.
 - Wiseman Prize 2023, Department of Computer Science and Technology, University of Cambridge.
 - DAAD AInet fellowship 2023.
 - Affiliated Postdoctoral Member, Clare College, University of Cambridge.
 - Member of Trinity College Postdoctoral Society, University of Cambridge.
 - Awarded ICWSM 2022 Virtual scholarship.
 - Awarded ICWSM 2021 scholarship for Underrepresented Groups Program.
 - Selected as a Rising Star in Data Science by the Center for Data and Computing (CDAC), University of Chicago.
 - Awarded 3rd prize at Ninth IDRBT Doctoral Colloquium IDC 2019.
 - Awarded Student Travel Grant to attend CODS COMAD 2019.
 - Grand Challenge Grade A winner at International Symposium on Embedded Devices (ISED) 2016 at IIT Patna.
 - Came 1st Runners Up in Startup Weekend, Kolkata (5-7th August, 2016).
 - Media Coverage of our Group in a local daily: <https://tinyurl.com/news-coverage-disarm>
 - Won 3rd Prize in Coding Competition at Techniche 2012 (IIT Guwahati Technical Fest Coding Event).
 - Attended NASSCOM Product Conclave-2016 as Student Researcher of ForkIT.
 - Rank of 3 in Master of Technology and Rank of 7 in Bachelor of Technology.
 - Secured 10/10 SGPA in M.Tech (3rd & 4th Semester).
 - Secured 10/10 SGPA in B.Tech (6th, 7th & 8th Semester).
 - GATE Qualified (2016) – Gate Score : 488 (Rank: 4370).
 - GATE Qualified (2015) – Gate Score : 441 (Rank: 6675).
 - Recipient of GATE Scholarship (2014) – Gate Score : 582 (Rank: 2139).
 - Passed ‘A’ level examination of National Cadet Crops (NCC).

- PROFESSIONAL SERVICES
- Public Library of Science ONE (PLOS One) (Reviewer).
 - ACM Web Conference (WebConf 2025) (Reviewer).
 - Special Interest Group on Information Retrieval (SIGIR 2025 Resource Track) (Reviewer).
 - Pacific-Asia Conference on Knowledge Discovery and Data Mining (PAKDD 2025) (Reviewer).
 - SIAM International Conference on Data Mining (SDM 2025) (Reviewer).
 - IEEE Open Journal of the Computer Society (Reviewer)
 - IEEE Building a Secure & Empowered Cyberspace (BuildSEC 2024) (Poster co-chair)
 - ACM Conference on Hypertext and Social Media (HyperText 2024) (Reviewer)
 - Workshop on Mobility in the Evolving Internet Architecture (MobiCom 2024) (Reviewer)
 - International Conference On Distributed Computing And Networking (ICDCN-SoCIEITY 2023) (Reviewer)
 - ACM International Conference on Web Search and Data Mining (WSDM 2024) (Reviewer)
 - Nature Scientific Reports (Reviewer).
 - Dewald Roode Workshop on Information Systems Security Research (DRW 2023) (Reviewer).
 - Special Interest Group on Information Retrieval (SIGIR 2023 Resource Track) (Reviewer).
 - Transactions on Asian and Low-Resource Language Information Processing (ACM TALLIP) (Reviewer).
 - International Joint Conference on Artificial Intelligence (IJCAI 2023) (Reviewer).
 - ACM Web Conference (WWW 2023) (Reviewer).
 - SIAM International Conference on Data Mining (SIAM 2023) (Reviewer).
 - Springer Machine Learning Journal (Reviewer).

- Cambridge Cybercrime Conference 2022 (Organizing committee)
- AAAI Conference on Artificial Intelligence (AAAI 2023) (Reviewer)
- ACM International Conference on Web Search and Data Mining (WSDM 2023) (Reviewer)
- International Conference on Neural Information Processing (ICONIP 2022) (Reviewer)
- ACM International Conference on Information and Knowledge Management (CIKM 2022 Short Paper) (Reviewer)
- Special Interest Group on Information Retrieval (SIGIR 2022 Resource Track) (Reviewer).
- Journal of Cybersecurity (Reviewer).
- IEEE Internet Computing (Reviewer).
- International Conference on Natural Language Processing (ICON-2021) (Reviewer).
- SIAM International Conference on Data Mining (SIAM 2022) (Reviewer).
- Springer Journal on Social Network Analysis and Mining (SNAM) (Reviewer).
- ACM International Conference on Information and Knowledge Management (CIKM 2021 Resource Track) (Reviewer).
- International Conference on Natural Language Processing (ICON-2020) (Reviewer)
- IEEE International Conference on Advanced Networks and Telecommunications Systems (IEEE ANTS 2020) (Reviewer)
- International Joint Conference on Web Intelligence and Intelligent Agent Technology (WI-IAT 2020) (Reviewer)
- AAAI Workshop on Combating online Hostile Posts in Regional Languages during Emergency Situation (CONSTRAINT 2021) (Web Chair & Reviewer)
- SIAM International Conference on Data Mining (SIAM 2021) (Reviewer).
- ACM International Conference on Information and Knowledge Management (CIKM 2020) (Reviewer).
- PAKDD Workshop on Data Science for Fake News (DSFN 2020) (Web Chair & Reviewer).
- Wiley Journal on Expert Systems (Reviewer).
- IEEE/WIC/ACM International Conference on Web Intelligence (WI 2019) (External Reviewer).
- International Conference on Secure Knowledge Management in Artificial Intelligence Era (SKM 2019) (External Reviewer).

EVENTS & WORKSHOPS ATTENDED

- Hidden Populations Workshop, University of Oxford (2022).
- Security and Human Behaviour (SHB), University of Cambridge (2021).
- Workshop on AI for Computational Social Systems (ACSS-20), IIIT Delhi (2021).
- Workshop on Combating Fraud Activities using Data Science (CoFad), IIIT Delhi (2020).
- Amazon Research Days, Bangalore (2019).
- Academic Research Summit 2019 on Data Science and AI, IIT Madras, Chennai (2019).
- ACM India Joint International Conference on Data Science and Management of Data (CoDS-COMAD) (2019).
- Workshop on AI for Computational Social Systems (ACSS-19), IIIT Delhi (2019).
- Winter School on Artificial Intelligence, IIIT Delhi (2019).
- Amazon Research Days, Bangalore (2018).
- Young Scientist Conference at India International Science Festival (2018).
- Workshop on Complex and Social Networks, IIT Kharagpur (2017).
- NASSCOM Product Conclave, Bangalore (2016).

TECHNICAL
SKILLS

Strongest Areas - Cybercrime, Data-driven Cybersecurity, Social Network Analysis, Data Mining, Applied Machine Learning, Web Mining, Natural Language Processing & System Programming.
Languages - Python, Java, PHP, Dart, C++, UNIX Shell, HTML, CSS, JS, Ajax, Perl
Tools - Git, Android Studio, Adobe Photoshop, JUnit, L^AT_EX
Databases - Firebase, MySQL, SQLite, MongoDB

PERSONAL
DETAILS

Date of Birth: 12th September, 1991
Gender: Male
Nationality: Indian